

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Linear Algebra for Machine Learning, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ~2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ~800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.

Vylet

May 2026 – Present | vyletdata.com

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels — manufacturers, SaaS tools, PE holding companies, geographic mismatches — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.

PROJECTS

SignalWeaver | *Python, LoRA*

github.com/Verdent06/SignalWeaver

Multi-signal financial research platform combining fundamentals, macro indicators, and news sentiment (research assistant; not investment advice)

- Lifted financial-sentiment classification accuracy from 81% to 96% by LoRA fine-tuning a quantized meta-llama/Meta-Llama-3.1-8B-Instruct model on 3,454 Financial PhraseBank entries, evaluated on a held-out test set.
- Built a linear regression combining fundamental, embedding-based sentiment, and LLM-derived sentiment signals into a composite return-prediction score; validated out-of-sample (3.39% R^2 , consistent with the low single-digit R^2 typical of real return-prediction literature) to confirm the model wasn't just fitting noise.

Granular Synthesizer Plugin | *C++, JUCE, VST3, AU*

github.com/Verdent06/granular-synth

Real-time audio DSP plugin in C++/JUCE — lock-free threading and release-ready VST3/AU binaries

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic `shared_ptr` swap so the UI thread never blocks `processBlock()`.

TECHNICAL SKILLS

Languages Python, C++, SQL, TypeScript
AI / ML Pandas, ONNX Runtime, Silero VAD, LangSmith
Frameworks Flask, JUCE, Angular, RxJS
Cloud & Infrastructure AWS (EC2, S3), Git